

# Quantum Topology-GRPO: A Variational Quantum Hierarchical Reinforcement Learning Algorithm with Global Topological Calibration for Intra-Graph Inter-Group Scenarios

Lanhaijian  
Hongqiao University

May 25, 2026

## Abstract

This paper proposes **Quantum Topology-GRPO**, a fully quantum-enhanced extension of the revised Topology-GRPO framework for graph-structured decision systems with inter-group coupling. The design strictly adheres to the core principle: *no intra-group critics, advantage computed purely from normalized rewards, with a global quantum value function reserved exclusively for topological calibration*. All classical modules are replaced with quantum-native implementations: inter-group communication via Bell-pair entanglement and quantum teleportation, topological advantage accelerated by quantum amplitude estimation, inter-group consistency enforced via SWAP-test state fidelity, and policy stability guaranteed by von Neumann entropy regularization. The algorithm preserves full alignment with GRPO fundamentals while unlocking quantum parallelism, entanglement-driven correlation modeling, and hardware compatibility for NISQ devices. We validate the framework on quantum chip layout, quantum-integrated scheduling, and quantum molecular generation tasks, with full theoretical derivation and open-source implementation provided.

## 1 Introduction

The revised Topology-GRPO framework eliminates intra-group critics and relies on reward normalization for advantage estimation, with a global value function retained only for cross-group topological validation. While this design achieves cleaner objectives and more stable training, classical implementations still face fundamental limitations in high-dimensional graph scenarios: limited expressivity for correlated group behaviors, linear scaling with neighborhood size, and inefficient global state representation.

Quantum computing provides a natural solution to these limitations:

- Quantum superposition enables exponential state representation and parallel evaluation.
- Quantum entanglement natively models inter-group correlations without explicit message overhead.
- Quantum amplitude estimation provides quadratic speedup for neighborhood reward statistics.
- Quantum state fidelity directly measures inter-group policy alignment without classical distance computation.

This paper upgrades the framework to Quantum Topology-GRPO with:

1. Fully quantum policy networks and message passing, with no intra-group critics.
2. Bell-pair entanglement for inter-group communication and quantum teleportation.
3. Quantum amplitude estimation for accelerated three-component topological advantage.
4. Global quantum value function  $V_{\text{global}}^q$  for topological calibration only.
5. SWAP-test fidelity constraints and von Neumann entropy regularization for policy stability.

## 2 Preliminaries and Quantum Formulation

We model the system as a weighted graph:

$$G = (V, E, W), \quad V = \bigcup_{g=1}^k V_g, \quad E = E_{\text{intra}} \cup E_{\text{inter}}.$$

In the quantum formulation:

- Nodes and groups are encoded into separable quantum registers  $|\psi_g\rangle$ .
- Inter-group edges are represented by pre-distributed Bell pairs:

$$|\Phi^+\rangle_{gh} = \frac{1}{\sqrt{2}} (|00\rangle_{gh} + |11\rangle_{gh}).$$

- The objective is to maximize the expected discounted return:

$$\max_{\pi_h^q, \{\pi_l^{g,q}\}} \mathbb{E} \left[ \sum_{g,t} \gamma^t R^{g,q} \right] \quad \text{s.t. inter-group quantum policy consistency.}$$

## 3 Quantum Topology-GRPO Framework

### 3.1 Core Design Principles

1. **No intra-group quantum critics:** no local value functions within any group.
2. **Advantage from quantum reward statistics only:** no TD learning, no local value fitting.
3. **Global quantum value for topological calibration only:**  $V_{\text{global}}^q$  exclusively for cross-group alignment.
4. **Quantum-native modules:** all components implemented with variational quantum circuits (PQCs).

### 3.2 Quantum State Encoding

Classical group states are mapped to quantum states via amplitude encoding:

$$|\tilde{\psi}_g\rangle = U_{\text{enc}}(s_g) \otimes U_{\text{msg}}(m_g) |0\rangle^{\otimes n},$$

where  $U_{\text{enc}}$  performs amplitude embedding for state information,  $U_{\text{msg}}$  performs phase encoding for inter-group messages, and  $n$  is the total number of qubits per group register.

### 3.3 Bell-Pair Quantum Inter-Group Message Passing

Inter-group communication is implemented via quantum teleportation over pre-distributed Bell pairs:

$$|\psi_{\text{msg}}\rangle_g = \text{Tr}_h \left[ U_{gh}(\theta_{gh}) |\tilde{\psi}_g\rangle \otimes |\tilde{\psi}_h\rangle \otimes |\Phi^+\rangle_{gh} \right],$$

where  $U_{gh}(\theta_{gh})$  is a parameterized two-qubit gate controlled by the coupling weight  $W_{gh}$ . The received message is fused with the local state via a quantum gating mechanism:

$$|\tilde{\psi}_g^{\text{enhanced}}\rangle = \text{QGateFuse} \left( |\tilde{\psi}_g\rangle, |\psi_{\text{msg}}\rangle_g \right),$$

implemented with RY rotation gates and quantum attention weighting.

### 3.4 Global Quantum Value for Topological Calibration

A global quantum value function is defined exclusively for cross-group credit alignment and topological validation:

$$V_{\text{global}}^q = \langle \psi_{\text{global}} | U_q^\dagger Z^{\otimes k} U_q | \psi_{\text{global}} \rangle,$$

where  $|\psi_{\text{global}}\rangle$  is the joint global quantum state, and  $Z^{\otimes k}$  is the Pauli-Z measurement operator. This value is not a learned critic and is only used to stabilize global advantage normalization.

### 3.5 Quantum Topological Advantage with Amplitude Estimation

The three-component advantage is computed purely from quantum reward statistics, accelerated by quantum amplitude estimation (Grover speedup from  $O(|\mathcal{N}(g)|)$  to  $O(\sqrt{|\mathcal{N}(g)|})$ ):

$$\begin{aligned}\hat{A}_{i,\text{quantum}}^g = & \alpha \cdot \frac{R_i^{\text{global}} - V_{\text{global}}^q}{\sigma^{\text{global}}} \\ & + (1 - \alpha) \cdot \frac{r_i^g - \bar{r}^g}{\sigma^g} \\ & + \beta \cdot \frac{r_i^g - \bar{r}^{\mathcal{N}(g)}}{\sigma^{\mathcal{N}(g)}}.\end{aligned}$$

All terms are derived from normalized rewards, with no intra-group value functions involved.

### 3.6 Quantum Inter-Group Consistency Loss

Classical L2 distance is replaced by quantum state fidelity, measured directly via SWAP-test:

$$\mathcal{L}_{\text{cons}}^q = - \sum_{(g,h)} W_{gh} \cdot F(\rho_g, \rho_h),$$

where  $\rho_g = \text{Tr}_{\text{env}}(|\pi_g\rangle\langle\pi_g|)$  is the reduced density matrix of group  $g$ 's policy output, and  $F$  is the quantum state fidelity.

### 3.7 Final Quantum Objective Function

The full loss function includes clipped policy loss, quantum consistency loss, KL divergence, and von Neumann entropy regularization to prevent policy collapse:

$$\begin{aligned}\mathcal{L}^{\text{Q-Topo-GRPO}} = & -\mathbb{E} \left[ \min \left( r_t^q \hat{A}_{i,\text{quantum}}^g, \text{clip}(r_t^q, 1 \pm \epsilon) \hat{A}_{i,\text{quantum}}^g \right) \right] \\ & + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}}^q \\ & + \beta D_{\text{KL}}^q(\pi_\theta \parallel \pi_{\text{ref}}) \\ & - \gamma H_{\text{von}}(\rho_\theta),\end{aligned}$$

where  $H_{\text{von}}(\rho) = -\text{Tr}(\rho \log \rho)$  is the von Neumann entropy,  $r_t^q$  is the quantum policy probability ratio, and  $D_{\text{KL}}^q$  is the quantum KL divergence.

## 4 Quantum Training Algorithm

---

#### Algorithm 1 Quantum Topology-GRPO Training

---

- 1: Initialize high-level quantum policy  $\pi_h^q$ , group quantum policies  $\{\pi_l^{g,q}\}$ , quantum message encoders
  - 2: Initialize global quantum value function  $V_{\text{global}}^q$  for topological calibration
  - 3: Pre-distribute Bell pairs for all inter-group edges  $E_{\text{inter}}$
  - 4: **while** not converged **do**
  - 5:   Construct graph  $G$  and split intra-group / inter-group edges
  - 6:   Collect parallel quantum trajectories across all groups
  - 7:   **for** each group  $g$  **do**
  - 8:     Perform Bell-pair quantum message passing to obtain enhanced state  $|\tilde{\psi}_g^{\text{enhanced}}\rangle$
  - 9:     Compute global, local, and topological neighborhood rewards
  - 10:    Compute quantum topological advantage  $\hat{A}_{i,\text{quantum}}^g$  via amplitude estimation (no intra-group critic)
  - 11:   **end for**
  - 12:   Evaluate clipped policy loss, quantum fidelity loss, KL divergence, and von Neumann entropy
  - 13:   Update quantum policy parameters via parameter-shift gradient descent
  - 14:   Decay  $\lambda_{\text{cons}}$  for curriculum learning
  - 15: **end while**
  - 16: **return**  $\pi_h^q, \{\pi_l^{g,q}\}$
-

## 5 Theoretical Properties

- **GRPO-aligned**: no intra-group critics, pure reward-based advantage estimation.
- **Quantum speedup**: quadratic speedup for neighborhood advantage via amplitude estimation.
- **Entanglement-native**: inter-group correlations modeled via Bell pairs, no explicit overhead.
- **Topologically consistent**: global quantum value ensures cross-group coherence.
- **NISQ-compatible**: all modules implemented with shallow variational quantum circuits.

## 6 Quantum-Enhanced Applications

### 6.1 Quantum Chip Layout

Groups: quantum modules, qubits, couplers; edges: entanglement connections; messages: quantum state fidelity,  $T_1/T_2$  decoherence times; consistency: cross-module gate fidelity matching.

### 6.2 Quantum Integrated Airline Scheduling

Groups: flights, crew, gates, quantum communication links; messages: superposed delay states, quantum key distribution status; consistency: phase synchronization and time-window alignment.

### 6.3 Quantum Molecular Generation

Groups: molecular scaffolds, functional groups; messages: valence quantum states, electronic correlations; consistency: quantum chemical symmetry constraints and chemical validity.

## 7 Implementation and Open Problems

Key implementation techniques:

- Quantum topology-aware gradient clipping for NISQ noise robustness.
- Quantum running mean/std normalization for reward scaling.
- Delayed entanglement synchronization for large-scale graphs.
- Curriculum learning for consistency strength annealing.

Open problems for future work:

1. Robustness under realistic NISQ device noise.
2. Optimal inter-group entanglement topology design.
3. Provable quantum advantage over classical Topology-GRPO.
4. Integration with quantum large language models (QLLMs) for high-level policy.

## 8 Conclusion

Quantum Topology-GRPO upgrades the revised Topology-GRPO framework to the quantum domain while strictly preserving its core principles: no intra-group critics, reward-based advantage estimation, and global value for topological calibration only. The framework leverages quantum entanglement, amplitude estimation, and state fidelity to achieve faster training, stronger inter-group coordination, and native compatibility with quantum hardware. It provides a theoretically grounded, hardware-compatible quantum hierarchical reinforcement learning framework for graph-structured decision systems.